

MAT 322 HW06

Aditya Pahuja

Due 04/01

Problem 1. Let S be the tetrahedron in $\mathbb{R}^3$ having vertices $(0, 0, 0)$, $(1, 2, 3)$, $(0, 1, 2)$, and $(-1, 1, 1)$. Evaluate $\int_S f$ where $f(x, y, z) = x + 2y - z$.

Solution. Let L be the linear transformation sending the tetrahedron T formed by the vectors $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ to S , specifically

$$L = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 1 \\ 3 & 2 & 1 \end{pmatrix}$$

Then

$$\int_{L(T)} f = \int_T (f \circ L) \cdot |\det(dL)|.$$

We can compute

$$(f \circ L)(x, y, z) = f(x - z, 2x + y + z, 3x + 2y + z) = (x - z) + 2(2x + y + z) - (3x + 2y + z) = 2x,$$

and, since L is linear, its derivative is equal to L , so $\det(dL) = \det L = -2$. Thus we wish to compute

$$4 \int_T x.$$

We can rewrite this as an iterated integral and compute

$$4 \int_{x=0}^1 \int_{y=0}^{1-x} \int_{z=0}^{1-x-y} x = \boxed{\frac{1}{6}}. \quad \square$$

Problem 2. Let U be the open set in $\mathbb{R}^2$ consisting of all x with $\|x\| < 1$. Let $f(x, y) = 1/(x^2 + y^2)$ for $(x, y) \neq (0, 0)$. Determine whether f is integrable over $U \setminus \{0\}$ and over $\mathbb{R}^2 \setminus \overline{U}$. If so, evaluate the integral.

Solution. For $n > 2$, let K_n be the annulus formed by deleting the open disk centered at $(0, 0)$ with radius $1/n$ from the closed disk centered at $(0, 0)$ with radius $1 - 1/n$. Then, clearly $\bigcup_n K_n = U \setminus \{0\}$. We can then integrate $\frac{1}{x^2 + y^2}$ over K_n via polar coordinates:

$$\int_{K_n} f = \int_{[0, 2\pi] \times [1/n, 1-1/n]} \frac{1}{r^2} \cdot r = 2\pi(\ln(1 - 1/n) - \ln(1/n)) = 2\pi \cdot \ln(n - 1).$$

This is unbounded as n grows large, so f is not integrable over U .

Similarly, for $n > 2$, let L_n be the annulus formed by deleting the open disk centered at $(0, 0)$ with radius $1 + 1/n$ from the closed disk centered at $(0, 0)$ with radius n ; we can see that $\bigcup_n L_n = \mathbb{R}^2 \setminus \overline{U}$. Furthermore, in polar coordinates,

$$\int_{L_n} f = \int_{[0, 2\pi] \times [1+1/n, n]} \frac{1}{r} = 2\pi(\ln(n) - \ln(1 + 1/n)) = 2\pi \cdot \ln(n^2/(n + 1)).$$

Again this is unbounded as n grows large, so f is not integrable over $\mathbb{R}^2 \setminus \overline{U}$. $\square$

Problem 3. Let B be the portion of the first quadrant in $\mathbb{R}^2$ lying between the hyperbolas $xy = 1$ and $xy = 2$ and the two lines $y = x$ and $y = 4x$. Evaluate $\int_B x^2 y^3$.

Solution. Transform $x \mapsto u/v$, $y \mapsto uv$ with $u, v > 0$, so that the region bounded by the hyperbolas and lines maps to the rectangle bounded between $u = 1$, $u = \sqrt{2}$, $v = 1$, and $v = 2$. Then the Jacobian for the transformation is

$$\begin{vmatrix} 1/v & -u/v^2 \\ v & u \end{vmatrix} = 2u/v.$$

Then $x^2 y^3 = u^5/v$, so, by Fubini's theorem, the integral equals

$$\int_R x^2 y^3 = 2 \int_1^2 \frac{1}{v^2} \cdot \int_1^{\sqrt{2}} u^6 = 2 \cdot \frac{1}{2} \cdot \frac{\sqrt{2}^7 - 1}{7} = \frac{8\sqrt{2} - 1}{7}.$$

□

Problem 4. (a) If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is of class C^1 , show that f is not one-to-one.

(b) If $f: \mathbb{R} \rightarrow \mathbb{R}^2$ is of class C^1 , show that f does not carry $\mathbb{R}$ onto $\mathbb{R}^2$. In fact, show that $f(\mathbb{R})$ contains no open set in $\mathbb{R}^2$.

Solution. (a) If $df = 0$ everywhere then f is constant. Otherwise let $p = (p_1, p_2)$ be a point at which (WLOG) $\frac{\partial f}{\partial y}(p) \neq 0$. Then, by the implicit function theorem, there exists a neighborhood $U \subseteq \mathbb{R}$ of p_1 and $V \subseteq \mathbb{R}$ of p_2 such that there is a C^1 function $g: U \rightarrow V$ satisfying $g(p_1) = p_2$ such that $f(x, g(x)) = f(p)$ for all $x \in U \times V$, proving that f isn't injective.

(b) We show that $f(\mathbb{R})$ has measure zero in $\mathbb{R}^2$.

Restrict f first to an interval $[a, b]$. Then, if $f(x) = (f_1(x), f_2(x))$ for C^1 functions f_1 and f_2 , by the mean value theorem, $f(x) - f(y) = (f'_1(\alpha_1), f'_2(\alpha_2))(x - y)$ for some $\alpha_1, \alpha_2 \in [a, b]$. Since $[a, b]$ is compact, f'_1 and f'_2 , being continuous, are bounded on $[a, b]$, hence $\frac{\|f(x) - f(y)\|}{|x - y|}$ is bounded for $x, y \in [a, b]$. This implies f is Lipschitz on $[a, b]$, so say $\|f(x) - f(y)\| \leq L|x - y|$ for all $x, y \in [a, b]$.

Let the points $a = t_0 \leq t_1 \leq \dots \leq t_n = b$ partition $[a, b]$ into intervals $[t_i, t_{i+1}]$ of equal length. Then, each set $f([t_i, t_{i+1}])$ can be covered with a cube of side length $L \cdot \frac{b-a}{2n}$, giving a covering of $f([a, b])$ with boxes of total volume at most

$$n \cdot \left(L \cdot \frac{b-a}{2n} \right)^d = \left(\frac{L \cdot (b-a)}{2} \right)^d \cdot n^{1-d},$$

which goes to zero as n grows arbitrarily large, implying that $f([a, b])$ has measure zero.

Now $f(\mathbb{R}) = \bigcup_{n=1}^{\infty} f([-n, n])$, which is a union of countably many sets of measure zero, hence also has measure zero.

This implies the result because all open sets do not have measure zero, on account of any open set containing an open ball, which clearly has nonzero measure. □